

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458195.8196 +/- 0.0035 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1168 +/- 0.0068
Transit depth (Rp/Rs)^2	1.36 +/- 0.16 [%]
Orbital Inclination (inc)	86.74 +/- 0.93
Airmass coefficient 1 (a1)	1.177 +/- 0.0088
Airmass coefficient 2 (a2)	-0.1442 +/- 0.0017
Scatter in the residuals of the lightcurve fit is	0.72 %
Best Comparison Star	None
Optimal Aperture	5.49
Optimal Annulus	16.53
Transit Duration (day)	0.099 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1168 $\pm$ 0.0068 vs 0.1326 (deviation 1.75 σ)
Duration fit vs published	0.099 d vs 0.1238 d (deviation 1.04 σ)
Mid-time O–C	10.0 min
Depth SNR	12.9
Residual scatter	0.72 %

Light curve

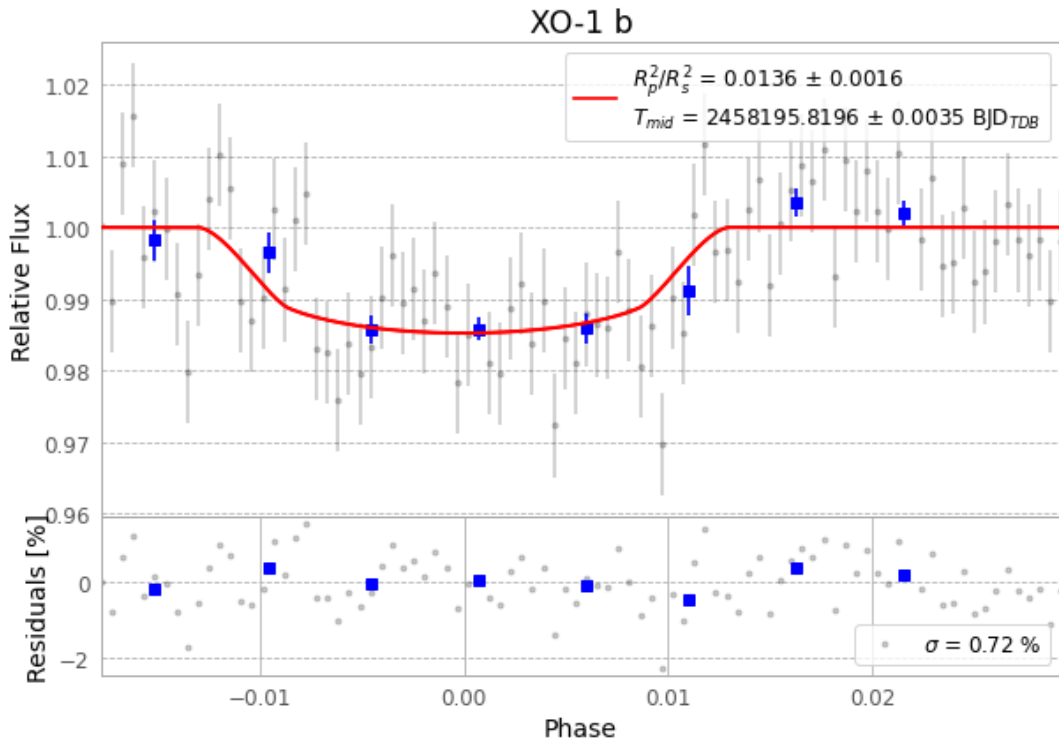


Figure 1 — FinalLightCurve_XO-1 b_2018-03-17.png (SHA-256 28872dc98b5cab0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [365, 149], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 968e2a2c08506740...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

92 FITS frames (60 MB), XO-1180318054812.FITS → XO-1180318102112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-XO-1-180318.log (0fd39a7ca727...), FinalLightCurve_XO-1 b_2018-03-17.png (28872dc98b5c...), FinalLightCurve_XO-1 b_2018-03-17.csv (568a8874ea4b...)

Compute resources

Wall time 265.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 03:00Z.